



**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 10	Team No- 10	Project Title- Interview Practice Platform – AI Resume Matching & Feedback
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1. Abstract

Job seekers today face two connected problems: over 75% of resumes are filtered out by ATS systems or recruiters before a human ever reviews them, and even candidates who land interviews often prepare using generic, one-size-fits-all question banks that don't reflect the actual role. Practicing alone also gives no structured feedback on answer quality, pacing, or delivery — the factors that most decide interview outcomes.

This project proposes an AI-powered Interview Practice Platform that addresses both gaps. Candidates upload their resume, which is parsed using spaCy to extract skills and experience. A sentence-transformers embedding model then scores the candidate's fit against a seeded job dataset, returning ranked, role-specific matches. Based on the matched job and identified resume gaps, an LLM (Groq/Gemini) generates targeted mock interview questions. Candidates respond via in-browser mic/camera capture (getUserMedia + MediaRecorder), and their spoken answers are transcribed using faster-whisper. The system then returns structured, LLM-based feedback on both content and delivery, storing sessions, transcripts, and scores in MongoDB for progress tracking over time.

The platform follows a three-tier architecture: a React frontend, a Node.js/Express backend handling auth and session orchestration, and a separate Python/Flask AI microservice for parsing, matching, question generation, and transcription — keeping AI logic isolated and swappable. The entire stack runs on free-tier and open-source infrastructure, making it accessible and low-cost to deploy, while directly closing the personalization and feedback gaps in current interview preparation tools.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used

- React.js — frontend
- Node.js + Express — backend/auth/orchestration
- Python (Flask) — AI microservice
- spaCy — resume parsing & entity extraction
- sentence-transformers — embedding-based job matching
- faster-whisper — speech-to-text transcription
- Groq / Gemini API — question generation & feedback
- MongoDB Atlas — session, transcript & score storage
- Git / GitHub — version control

Guide Name and Signature (remarks if any) _____